Fiscal Year:	FY 2021	**Task Last Updated:**	FY 06/05/2025
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Project Title:	Concurrent Flame Spread Modeling Using Flamelet Generated Manifolds in Micro-Gravity with Comparison to BASS Experiments Using Two-Color Tomography		
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Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
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Task Description:

One of the largest modeling challenges in understanding concurrent flame spread is the coupling between thermal and mass transport processes along with chemical kinetics near the solid-vapor interfaces. Since explicitly resolving all these processes in 3D at engineering scales is impossible even with today's most advanced supercomputers, a modeling or scaling methodology must be introduced to correlate near-surface behavior to far-field flow dynamics. Current modeling approaches for defining this coupling often rely on the superposition of turbulence and chemistry models that are not theoretically or mathematically self-consistent, e.g., use of a near-wall turbulence model for shear-stress, coupled with a Newton's law of cooling for heat transfer, coupled with a simplified ad-hoc Arrhenius expression for the surface burning rate. Predictions are often qualitatively correct at best and do not include the details of important intermediate chemistry steps which define pollutants. A new modeling approach is therefore desirable which, at a minimum, includes the detailed coupling of all relevant processes in the near-wall region.
The objective of the proposed research is to explore newly developed flamelet generated manifold (FGM) modeling approaches for use in concurrent flame spread modeling. Central to this approach is a newly developed unsteady FGM (UFGM) modeling approach for reacting interfaces which maps the reacting state space into lower dimensional manifolds. The UFGM allows for affordable calculations of multidimensional simulations of burning phenomena. Fully coupled numerical simulations of a subset of the Burning and Suppression of Solids (BASS) experiments will be conducted using a computational framework developed over the last 15 years by the Principal Investigator. The framework allows for fully coupled simulations of fluid-solid response – specifically designed for charring and ablating materials. In the proposed effort, the use and validation of UFGM will be explored for use in prediction of flame spread from NASA's BASS and BASS-II experiments. The imagery from these experiments will be post-processed using newly developed two-color tomography techniques for digital single lens reflex (DSLR) cameras so 3D soot and temperature fields may be determined and compared to modeling predictions.

The appeal of using the BASS series for validation is the absence of buoyancy forces, allowing for unambiguous assessment of the UFGM modeling to predict concurrent flame spread in complex geometries, e.g., sphere and end rod configurations. Simulations of flat, spherical, and rod Polymethyl methacrylate (PMMA) samples will be conducted, along with the cylinder geometry of wax. Specific metrics are identified for the comparisons which include flame temperature, soot volume fraction, flame geometry, flame spread rate, etc. A particularly interesting phenomenon of interest to explore with the model is to see if it can reproduce the `Goldilocks' flammability zone discussed recently by Olson and Ferkul. These comparisons will be conducted in conjunction with an on-going National Science Foundation (NSF) funded project in exploring UFGM for upward flame spread so relative comparisons of model agreement in terrestrial and non-terrestrial settings can be assessed. The long-term impact of developing the UFGM modeling approach is the ability to screen new material flammability limits which may be used in future spacecraft. In addition, the UFGM can be used as a subgrid scale model (SGS) for Large Eddy Simulations (LES) of fire to explore potential hazardous scenarios on spacecraft.

S. L. Olson and P. V. Ferkul. Micogravity flammability boundary for PMMA rods inaxial stagnation flow: Experimental results and energy balance analyses. Combust. and Flame, 180:217-229, 2017.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The focus of this research was to explore newly developed flamelet generated manifold (FGM) modeling approaches for use in opposed flow flame spread modeling in microgravity. The FGM allows for affordable calculations of multidimensional simulations of burning phenomena. A goal of the work is to use data from the NASA Burning and Suppression of Solids (BASS) and BASS-II experiments to validate the detailed modeling simulation tools developed. Specifically, data related to solid fuel flammability limits and video data for use with two-color pyrometry techniques was the main interest.
The bulk of the research effort was on the development of 1D radiative flamelets that allow the incorporation of detailed chemistry (gas + soot) and radiation heat transfer. The flamelets are shown to capture the flammability bounds from BASS experiments through a dissipation matching exercise, and useful analytical limits using thin-flame theory are also derived. To test the use of these flamelets for use in opposed flame spread mirogravity, a detailed 2D flamespread model was developed that includes state-of-the-art chemistry and radiation heat transfer transport. Comparisons of flamelet models via the construction of FGMs are compared to the detailed 2D simulations in an a priori error study. An interesting finding from this activity was finding creative means to span the composition space from the 2D simulations using unsteady extinguishing flames in both the radiative quenching and blow-off limits. Overall errors between flamelet solutions and the simulations are quite low, with observed differences in regions near flame attachment where multi-dimensional wall heating processes dominate.

Two-color pyrometry advances from the project: The main advance in this topic was the creation of a two-color flamelet manifold. The fundamental finding from this work was the discovery of a one-to-one mapping between flame strain (dissipation rate) and the ratio of red-to-green color intensity from a digital single lens reflex (DSLR) camera. The mapping was determined using virtual two-color pyrometry through the 1D radiative flamelets and used to map flamelets to actual thermocouple (TCP) measurements of flames. This allowed an approximate 3D flame reconstruction using only two camera side views. Knowing the flame hull and cross-sectional composition allowed for full 3D ray tracing to determine flame-to-surface radiative heat flux, and showed excellent agreement to independent measurements using Schmidt-Boelter heat flux gauges. Since challenges were encountered acquiring the actual NASA camera used in the BASS experiment due to COVID related restrictions, independent ground based experiments were instead conducted at the University at Buffalo (UB), using a camera with similar specifications to demonstrate proof-of-concept.

The focus of the final year of the effort was on: 1) validating the two dimensional opposed flow flame spread model with detailed chemistry and radiation using experimental data and previous modeling efforts on the flame spread rate; 2) developing a simple one dimensional flamelet problem that include the radiative and conjugate heat and mass transfer coupling at the fuel surface for use in generating newly developed coupled radiative FGMs (CR-FGMs) and exploring the flammability bounds for solid fuel diffusion flames in microgravity; 3) the development of an augmented thin flame theory including radiative and finite rate effects to explore the governing physical properties affecting the flammability bounds; and 4) construction and application of the CR-FGM's to the two dimensional opposed flow flame spread model. The results from these activities were compiled and submitted for publication. [Ed. Note: See Cumulative Bibliography.]

Bibliography Type:	Description: (Last Updated: 06/05/2025)
Articles in Peer-reviewed Journals	Budzinski K, DesJardin PE. "Radiative flamelet generated manifolds for solid fuel flame spread in microgravity." Combustion and Flame. 2023 Oct 1;256:112939. https://doi.org/10.1016/j.combustflame.2023.112939 , Oct-2023
Articles in Peer-reviewed Journals	Budzinski K, DesJardin PE. "Theoretical estimates of flammability bounds for thin condensed fuel diffusion flames in microgravity using detailed models of chemistry and radiation." Combustion and Flame. 2023 Sep 1;255:112910. https://doi.org/10.1016/j.combustflame.2023.112910 , Sep-2023